WEEK 1 : Project Idea Finalization & Team Registration

Section A – Group Information & Monitoring Log

Field	Details
Week	1
Milestone	Project Idea Finalization & Team Registration
Group ID / Team Name	
Student Names, Roll Numbers, Mobile No. and Email ID:	1. Anuj Kumar – 2310991699 anuj1699.be23@chitkara.edu.in +91-9350773108 2. Arpit Madan – 2310991707 arpit1707.be23@chitkara.edu.in +91-9467702434
Project Supervisor/ Guide Name with email and Mobile Number	
Initial Comments by Guide (if any)	
Guide Approval	☐ Approved ☐ Revisions Needed
Resubmission Date (if any)	

Section B – Project Idea Submission (To be filled by Student Team)

Field	Student Input
1. Domain/Field	Artificial Intelligence & Machine Learning, Computer Vision, Cybersecurity, Digital Forensics
2. Project Title	DeepTrace: Explainable Deepfake and AI-Generated Image Detection System
3. Problem Statement	Deepfake and AI-generated images are increasingly used for misinformation, fake identities, and online fraud. It is difficult for normal users to verify whether an image is real or manipulated. DeepTrace aims to build a web-based system that detects real, deepfake, and AI-generated images using deep learning. It will provide a confidence score, risk level, and simple explanation to help users verify image authenticity.

4. Inputs	The system will require image files uploaded by users, such as face images, suspected deepfake images, or AI-generated images. It may also use image metadata such as file type, resolution, creation details, compression information, and optional user details for maintaining analysis history.
5. Outputs	The system will provide a real/fake prediction, confidence score, risk level, metadata analysis, visual forensic indicators, suspicious-region heatmap, analysis history, and a downloadable PDF report containing the final result and explanation.
6. Functionalities (Minimum 6)	1. User registration and login 2. Image upload for deepfake or AI-generated image detection 3. Image preprocessing and face detection 4. Deep learning based real/fake classification 5. Confidence score and risk-level generation 6. EXIF/metadata analysis 7. Error Level Analysis for forensic checking 8. Explainable heatmap for suspicious regions 9. Analysis history dashboard 10. Downloadable forensic report
7. Use Cases	Students, teachers, journalists, social media users, and cyber forensic teams can use the system to check whether an image is real, deepfake, or AI-generated. A user uploads a suspicious image, and the system analyzes visual patterns, metadata, and model prediction to generate a risk score and explanation. The system can help reduce misinformation, fake identity misuse, and digital media fraud.
8. Tools & Technologies	*Python, FastAPI, React.js, Tailwind CSS, MongoDB, JWT Authentication, PyTorch or TensorFlow, OpenCV, PIL, MTCNN/face detection, Grad-CAM/heatmap visualization, EXIF metadata tools, ReportLab or jsPDF for PDF report generation, GitHub, Render/Vercel for deployment.*
9. Expected Team Roles (follow same order as per Student Names & Roll Numbers)	- Member 1: Anuj Kumar - 2310991699 - Team Lead, Backend Development, ML Model Integration, Testing - Member 2: Arpit Madan - 2310991707 - Frontend Development, UI Design, Documentation, Presentation
10. Methodology	The project will follow a client-server architecture. The frontend will allow users to upload images and view detailed analysis results. The backend will preprocess the image, detect faces, extract forensic metadata, and pass the image through a deep learning model for real/fake classification. The final output will combine model

	confidence, metadata signals, and explainability heatmaps to generate a risk score and PDF report.

Faculty Feedback / Notes

(To be filled during weekly review session or submitted in shared tracker)
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